MIL-F-43514B
30 September 1985
SUPERSEDING
MIL-F-43514A
1 June 1971

MILITARY SPECIFICATION

FASTENER, PLASTIC, FOR EQUIPAGE ITEMS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the requirements for one type of plastic fastener for equipage items. The fastener consists of four separate molded parts: a latch, a latch receptacle, a locking clip for attaching the latch, and a locking clip for attaching the latch receptacle.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-392	-	Plastic Molding Material, Acetal, Injection and Extrusion
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8465

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

8-2-225 - Fastener, Plastic, for Equipage Items
8-2-389 - Fastener, Plastic, for Equipage Items; Attaching Tools

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

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3. REQUIREMENTS

3.1 Design and dimensions. The fastener shall be composed of the following parts: latch, latch receptacle, locking clip for latch, and locking clip for latch receptacle. The design and dimensions of the fastener parts shall be in accordance with Drawing 8-2-225.

* 3.2 Materials. All parts shall be made from acetal molding compound conforming to Type I, class 2 of L-P-392 except that the electrical property requirements shall not apply. Clean, unburned plastic material in the form of imperfect parts, sprues, runners or other scraps of the same composition, produced in the molding or finishing operation, may be reground and mixed with the material. Recycled material may be used (see 6.4). The molding material may contain a surface lubricant.

3.3 Color and finish. The color of the finished fastener shall be black. The fastener parts shall have a lusterless or dull finish on all surfaces except as indicated on Drawing 8-2-225. The finish shall be produced as a result of the molding operation and not by application of any protective coating, lacquer, or other materials.

3.4 Marking for indentification. The fastener shall be marked by an embossed molded-in process with the manufacturer's name or trademark and the year of manufacture. The marking shall be located on any surface except those surfaces which are outermost when the fastener is in use. The letter size shall be approximately 3/32 inch in height and the characters shall be raised approximately 0.007 inch.

3.5 Performance.

3.5.1 Strength of setting pinch. The clinch of the latch and its clip and the receptacle and its clip shall each withstand a minimum pull of 120 pounds when tested as specified in 4.3.2.

3.6 Workmanship. The parts shall be clean, smooth, and free from cracks, dirt, flash, and surface defects such as splay and jetting marks. All gates shall be trimmed flush with the molding surface. The posts on the latch and latch receptacle shall be fully formed and free from any imperfection which may interfere with their locking engagement when assembled to the locking clips.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Certificates of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Inspection of materials and components. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase documents. A certificate of compliance shall be furnished with each shipment or lot stating that scrap molding materials, if used, were limited to those specified in 3.2.

4.2.2 Examination of the end item. The fasteners shall be examined in accordance with 4.2.2.1 and 4.2.2.2. The lot size shall be expressed in units of one complete fastener. The sample unit shall be one complete fastener.

4.2.2.1 Visual examination of the end item. The fasteners shall be examined for defects listed in table I. The inspection level shall be I and the acceptable quality level (AQL) shall be 4.0, expressed in defects per hundred units.

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TABLE I. Examination of end item

Examine	Defect
Design and construction	Not in accordance with Drawing 8-2-225
Color and finish	Not black Not dull or lusterless where specified Not uniform Any evidence of coating
Marking	Omitted; incorrect; illegible; improper size, location, or method of application
Workmanship	Any part broken, cracked, chipped, distorted, twisted, or out of shape Any dirt or flash Any deep scratch or gouge Gates not trimmed as specified Surface not smooth Any pit, void, crazing, air pocket, blister, or imbedded foreign matter that will affect serviceability Evidence of splay or jetting marks Latch and latch receptacle do not mate

4.2.2.2 Dimensional examination. The fasteners (all parts) shall be examined for defects in dimensions. Any dimension that varies from that specified in Drawing 8-2-225 shall be classified as a defect. The inspection level shall be S-2 and the AQL shall be 4.0 expressed in percent defective.

4.2.3 Packaging inspection. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. The lot size shall be the number of shipping containers in the inspection lot. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

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TABLE II. Packaging inspection

Examine	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing Any component damaged
Workmanship	Inadequate application of components such as: incomplete closure of container flaps, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Contents	Number per interior box or shipping container of any of the four component parts of the fastener (latch, latch receptacle, locking clip for latch and locking clip for latch receptacle) is more or less than specified.

* 4.2.4 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list in table III. The lot size shall be the number of palletized unit loads in the end item inspection lot. The sample unit shall be one palletized unit load fully packaged. The inspection level shall be S-1 and the AQL shall be 6.5, expressed in terms of defects per hundred units.

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TABLE III. Examination for palletization

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application

4.3 Testing of the end item. The fastener shall be tested for strength of the clinch in accordance with 4.3.2. The lot size shall be expressed in units of one complete fastener. The sample unit for this test shall be one complete fastener. The inspection level shall be S-2, and the lot shall be unacceptable if any sample unit fails to meet the requirement specified in 3.5.1. All test reports shall contain the individual values utilized in expressing the final result.

4.3.1 Test conditions. Test specimens shall be conditioned at 73.5° ± 2°F and 50 percent ± 4 percent relative humidity for 24 hours prior to the test, and the test shall be performed under these conditions.

4.3.2 Procedure for testing strength of setting pinch (clinch). Assemble the latch receptacle and its locking clip with two pieces of brass strip (0.010 inch thick) stock, as shown in Drawing 8-2-225. The brass strips shall be pre-punched before assembling. Clinch the posts of the latch receptacle by hot or cold heading by the same method to be used in the manufacture of the equipage items on which the fastener will be attached. The top of the clinched posts shall be below the surface of the locking clip. The specimen for testing the clinch of the latch shall be prepared in the same manner as that for the latch receptacle except that the latch shall be cut off one inch from the top and the top part (including the posts) of the latch shall be used

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in the test. The "setting pinch," after heading, for the latch with latch locking clip and the latch receptacle with latch receptacle locking clip under this test shall be 0.310 ± 0.004 inch and 0.625 ± 0.004 inch, respectively. The "setting pinch" is the measurement at which fastener units are correctly attached, and is obtained by adding the compressed thickness of any given foundation (in this case, brass strips) to the "basic pinch." The "basic pinch" of a fastener unit (latch with latch locking clip or latch receptacle with latch receptacle locking clip) is the measurement, made by micrometer, from the exposed surface of the latch or latch receptacle to over the rolled ends of the posts as these parts are brought into contact with their locking clips in the normally applied relationship. The "basic pinch" for the latch with locking clip and the latch receptacle with locking clip is 0.290 inch and 0.605 inch, respectively. The die that clinches the posts shall be a double tested rolling die, common to both the latch and latch receptacle (see 6.3). The detail of the working area of the die is shown on Drawing 8-2-389. With the test assembly in a horizontal position, bend the two ends of one of the brass strips upward so that they touch and clamp these ends together in the upper clamp of a tensile testing machine. Bend the two ends of the other brass strip downward so that they touch and clamp these ends in the lower clamp of the tensile tester. The distance between the jaws at the beginning of the test shall be four inches and the rate of jaw separation shall be twelve inches per minute. Report to the nearest pound the force indicated on the machine at the time of rupture of the assembly. Any specimen rupturing (completely separating) at a force below 120 pounds shall be considered to have failed the test.

* 5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each of the four plastic fastener components (latch, latch receptacle, locking clip for latch, and locking clip for latch receptacle) shall be unit packed in quantities of 1000 in individual fiberboard boxes conforming to style RSC, type CF (variety SW), class domestic, grade 175 of PPP-B-636. Each fiberboard box shall be securely closed with 2-inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45. Indentical size fiberboard boxes shall be used for the unit packing of the respective components. Voids in the fiberboard boxes shall be filled with cushioning material to immobilize the contents.

5.1.2 Commercial. The plastic fasteners shall be preserved in accordance with ASTM D 3951.

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5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. One thousand complete plastic fasteners (consisting of 1000 each of the latch, latch receptacle, locking clip for latch, and locking clip for latch receptacle) preserved as specified in 5.1 shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.3.

5.2.2 Level B packing. One thousand complete plastic fasteners (consisting of 1000 each of the latch, latch receptacle, locking clip for latch, and locking clip for latch receptacle) preserved as specified in 5.1 shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.3.

5.2.2.1 Weather-resistant containers. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.3.

5.2.3 Commercial packing. Plastic fasteners preserved as specified in 5.1 shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), fasteners packed as specified in 5.2 shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. If the container is of a size that does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

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6. NOTES

6.1 Intended use. The fastener is intended to be used as a closure device for equipage items such as ammunition pouches, carrier for collapsible entrenching tool, and collapsible canteen covers. The fastener is intended only for attachment to materials having a compressed thickness of 0.010 inch minimum and 0.050 inch maximum.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).
c. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
d. When palletization is required (see 5.3).

6.3 Tools for attaching the fasteners are shown on Drawing 8-2-389. Tools of this design have been found to attach the fastener parts satisfactorily. These tools were designed to fit into Scovill Manufacturing Company's Model No. 130-L-D attaching machine (foot operating). Accordingly, changes may be made in the design of the tooling in order to adapt it to supplier's attaching equipment except for the design of the working area of the double treated rolling die.

* 6.4 Recycled material. The contractor is encouraged to use recycled material when practical, provided that it meets the requirements of this document (see 3.2).

6.5 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-43514B	Fastener, Plastic, For Equipage Items

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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